CENG401 –FİNAL REPORT

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1. Project Information

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Research Title:	Artificial Intelligence And Image Processing-Based For Food Intake Measurement And Consumption Pattern Analysis in University Dining Halls		
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2. Brief Summary of the Project Scope, Objectives, and Current Progress

This project addresses the problem of food waste in large-scale university dining halls by proposing an artificial intelligence–based system for measuring real food consumption. In current cafeteria operations, meal production decisions are typically made without objective, dish-level consumption data, leading to systematic overproduction, increased food waste, and inefficient resource utilization. The absence of automated and scalable measurement mechanisms prevents data-driven menu planning and waste reduction.

The objective of this project is to design and implement a computer vision–based system capable of recognizing cafeteria dishes and supporting consumption estimation. The proposed approach relies on deep learning techniques, particularly object detection–based deep learning techniques, to analyze images captured before and after consumption. Reference images of meals are collected at serving time, while post-consumption tray images are captured at the tray-return stage to enable comparative analysis and support future consumption estimation.

As part of the project, a prototype dataset was constructed using images collected in a real dining hall environment. A deep learning model based on the YOLO architecture was trained using transfer learning to perform dish recognition and localization under realistic conditions, including irregular food shapes and lighting variations. Experimental evaluations demonstrate that the system can reliably identify dishes and approximate leftover quantities with acceptable accuracy for prototype-level decision support.

In addition to the core AI pipeline, a prototype application interface was designed to visualize dish-level consumption statistics, waste percentages, and temporal trends for cafeteria administrators. Overall, the project demonstrates the feasibility of an automated, AI-driven monitoring system that can support data-driven menu optimization and contribute to reducing food waste in institutional dining environments.

3. Problem Definition and Project Objectives

3.1 Problem Definition

University dining halls serve large numbers of meals daily and operate under strict time and cost constraints. Despite this scale, most cafeterias lack objective, dish-level data that describe how much of each served meal is actually consumed. Meal production decisions are therefore based largely on intuition or historical habits rather than measurable consumption patterns. As a result, unpopular dishes may continue to be overproduced, leading to excessive food waste, unnecessary operational costs, and negative environmental impact.

Food waste in institutional dining environments primarily occurs at the post-consumption stage, where significant portions of meals remain uneaten on returned trays. Existing waste measurement approaches, such as manual observation, weighing leftovers, or periodic surveys, are labor-intensive, subjective, and impractical for continuous daily use. Moreover, these methods fail to provide fine-grained, dish-specific insights that could support targeted menu optimization.

While recent advances in computer vision and artificial intelligence have enabled food recognition and nutritional analysis, most existing systems focus on personal dietary tracking rather than large-scale institutional monitoring. There is currently no widely adopted automated solution capable of continuously measuring dish-level consumption in real cafeteria environments without disrupting operations. This gap prevents dining hall administrators from making data-driven decisions aimed at reducing waste and improving resource efficiency.

3.2 Project Objectives

The main objective of this project is to develop an AI-driven system that automatically measures food consumption and waste in a university dining hall using computer vision techniques.

The specific objectives of the project are as follows:

1. To construct a representative image dataset consisting of reference meal images and post-consumption tray images captured in a real dining hall environment.

2. To design and train a deep learning–based object detection model capable of recognizing cafeteria dishes and localizing leftover food regions.
3. To estimate dish-level consumption by comparing served and remaining food portions using image-based analysis.
4. To evaluate the performance of the proposed system under realistic conditions, including irregular food shapes and environmental variability.
5. To develop a prototype application interface that visualizes consumption statistics and supports data-driven menu planning and waste reduction.

Together, these objectives aim to demonstrate the feasibility of an automated, scalable, and non-intrusive solution for institutional food waste monitoring.

4. Related Work and Its Relation to This Project

Research on AI-based food analysis has grown rapidly in recent years, mainly driven by applications such as food recognition, calorie estimation, and personal dietary monitoring. However, most existing studies focus on individual-level nutrition tracking rather than institutional-scale waste measurement. This section summarizes the most relevant work and explains how this project extends current approaches.

4.1 Food Recognition and Classification

Early work in computer vision for food analysis largely focused on recognizing food categories from images. A notable contribution is the Food-101 dataset, introduced by Bossard et al., which demonstrated that convolutional neural networks can classify visually similar food items effectively. While classification-based approaches establish a foundation for recognizing what food appears in an image, they do not provide information about portion size, leftover quantity, or real consumption in a cafeteria setting. For institutional waste monitoring, recognizing "what the dish is" is necessary but not sufficient.

4.2 Segmentation-Based Food Analysis

Later studies moved beyond classification toward localization and segmentation. Object detection models such as YOLO enable localization of food items and have become widely adopted for real-time applications. This capability is critical for scenarios where multiple dishes appear together, overlap, or have irregular boundaries. However, most localization and segmentation research targets clean images or controlled conditions and typically aims at identifying food items rather than measuring consumption after eating.

4.3 Portion Size and Volume Estimation

Another line of research focuses on estimating portion size, volume, or calories from images. Im2Calories (Meyers et al.) explored calorie estimation from single-view images by combining recognition with geometric cues. Similarly, Puri et al. investigated recognition and volume estimation using mobile devices. These studies demonstrate that quantity estimation is technically feasible, but they primarily address personal meal tracking and are not designed for high-throughput institutional environments. In addition, single-image methods struggle to account for partial consumption and variability in serving sizes.

4.4 Plate Waste Measurement in Institutional Settings

In institutional contexts, plate waste analysis has often relied on manual or semi-automated approaches. Swanson used digital photography as a tool to measure meal consumption, but the evaluation typically required human interpretation and was labor-intensive. Traditional methods such as manual weighing or observation can provide accurate waste measurements, yet they do not scale well for daily cafeteria operations and usually fail to produce continuous dish-level analytics.

4.5 Gap in Existing Work and Contribution of This Project

The literature reveals three key gaps that this project targets:

1. **Lack of scalable before–after consumption measurement:**
   Many systems analyze only a single image and cannot reliably estimate what was consumed.
   Our approach uses a dual-stage strategy: reference images at serving time and tray-return images after consumption.

2. **Limited focus on institutional waste reduction:**
   Prior work is largely oriented toward personal nutrition monitoring.
   Our project focuses on university dining halls and aims to deliver dish-level analytics for operational decision-making.

3. **Insufficient robustness for real cafeteria tray conditions:**
   Real trays include messy leftovers, occlusions, and lighting variations that degrade model performance.
   Our system design explicitly targets these real-world constraints through segmentation-based analysis and iterative refinement using a prototype dataset collected in the dining hall environment.

Overall, this project builds on established deep learning foundations (food recognition and instance segmentation) and extends them toward an institutional decision-support system that estimates dish-level consumption and waste, enabling data-driven menu planning and sustainability improvements.

5. Methods, Tools, Technologies, and Techniques

This project adopts an end-to-end system architecture that combines computer vision, deep learning, and data analytics to estimate food consumption and waste in a university dining hall. The methodology is designed to operate under real cafeteria conditions while remaining non-intrusive and scalable.

5.1 System Architecture Overview

The proposed system consists of four main components:

1. **Data Acquisition Layer** – image capture at serving time and tray-return stage
2. **Preprocessing Layer** – image normalization and enhancement
3. **AI-Based Analysis Layer** – dish recognition, localization, and leftover estimation support
4. **Application Layer** – visualization and decision-support interface

This modular architecture allows each component to be developed, evaluated, and improved independently.

5.2 Data Acquisition

Two image acquisition stages are employed. Reference images are captured at the serving line to represent fully served meals. Post-consumption images are captured at the tray-return area after students finish eating. Images are collected using standard RGB cameras positioned at fixed angles to ensure consistency while minimizing interference with cafeteria operations.

This dual-stage data acquisition strategy enables comparative analysis between served and consumed food portions. While segmentation-based approaches are powerful, this project adopts a YOLO-based detection framework to balance accuracy and computational efficiency in real cafeteria settings.

5.3 Data Preprocessing

To improve robustness under varying environmental conditions, the following preprocessing techniques are applied:

- Image resizing and normalization

- Illumination correction to reduce lighting variability
- Contrast enhancement for improved feature visibility
- Data augmentation techniques such as rotation, brightness adjustment, and contrast shifting

These steps help reduce noise and improve generalization of the deep learning model.

5.4 Deep Learning Model and Training

The core analytical component of the system is a YOLO-based object detection model, selected due to its efficiency and suitability for real-time applications.

Key model characteristics include:

- Backbone architecture: ResNet-50
- Transfer learning using pre-trained weights on large-scale datasets
- Fine-tuning using the constructed cafeteria dataset

Preliminary training experiments were conducted to evaluate the feasibility of dish detection. Supervised learning is applied using manually annotated images.

5.5 Leftover Estimation and Consumption Analysis

Leftover estimation is performed by analyzing the detected food regions in post-consumption images and comparing them with reference images. The system estimates relative consumption by approximating the remaining food area and, where applicable, incorporating geometric and statistical assumptions for volume estimation.

Dish-level consumption metrics are then aggregated over time to identify patterns such as frequently under-consumed meals or high-waste days.

5.6 Application Interface and Visualization

A prototype application interface has been designed to present system outputs in an accessible and actionable format. The dashboard visualizes key indicators such as:

- Dish popularity rankings
- Leftover percentage per dish
- Daily and weekly consumption trends
- Alerts for consistently high waste rates

The interface is designed to support cafeteria administrators in making data-driven decisions regarding menu planning and portion control.

5.7 Tools and Technologies

The system is implemented using widely adopted tools and frameworks:

- **Programming Language:** Python
- **Deep Learning Frameworks:** PyTorch
- **Computer Vision Libraries:** OpenCV
- **Model Architecture:** YOLO
- **Data Annotation Tools:** Polygon-based labeling tools
- **Application Layer:** Web-based prototype dashboard

These technologies collectively support rapid prototyping, experimentation, and future scalability.

6. Documentation of All Tasks Completed Throughout the Project

This section documents the tasks that have been carried out throughout the project to date. The project is currently in an active development phase, with dataset construction and model training ongoing. The completed tasks demonstrate steady progress toward the final system implementation.

6.1 Project Planning and Research

- Identification of food waste as a critical problem in large-scale university dining halls.
- Definition of project scope, system requirements, and research objectives.
- Comprehensive literature review on food recognition, object detection, food waste measurement, and AI-based consumption analysis.
- Analysis of existing approaches and identification of limitations related to institutional-scale deployment.

6.2 Dataset Design and Ongoing Data Collection

- Design of a dual-stage data acquisition strategy consisting of:

    - Reference images captured at serving time
    - Post-consumption tray images captured at the tray-return stage
- Initiation of data collection in a real university dining hall environment under varying lighting, tray orientation, and food presentation conditions.
- Organization of collected images into structured datasets suitable for object detection and consumption analysis.
- Ongoing expansion of the dataset to increase class diversity and sample balance.

6.3 Data Annotation and Preprocessing

- Manual annotation of food images using bounding boxes compatible with YOLO-based object detection.
- Definition of dish classes and verification of annotation consistency across different food categories.
- Application of preprocessing techniques such as image resizing, normalization, and basic data augmentation to improve model generalization.
- Continuous refinement of annotations as new data is collected.

6.4 Model Development and Training (Ongoing)

- Selection of a YOLO-based object detection model for food recognition and localization due to its efficiency and suitability for real-time applications.
- Configuration of the training pipeline using transfer learning with pre-trained YOLO weights.
- Initiation of model training on the constructed dataset and preliminary performance analysis using standard evaluation metrics.
- Iterative adjustment of model parameters, confidence thresholds, and training settings based on observed results.
- Ongoing training to improve robustness against visually similar dishes and challenging tray conditions.

6.5 Prototype Implementation and Preliminary Evaluation

- Integration of data preprocessing and YOLO-based inference into a prototype analysis pipeline.
- Generation of preliminary outputs such as detection visualizations, confusion matrices, and F1–confidence curves.
- Qualitative evaluation of detection results through visual inspection of predicted bounding boxes on sample images.
- Identification of challenging classes and scenarios to guide further data collection and model refinement.

6.6 Application Interface Design (Prototype Stage)

- Design of a conceptual dashboard interface to visualize detection outputs and dish-level statistics.
- Definition of key interface components including detected dish labels, confidence scores, and summary statistics.
- Preparation of prototype visualizations to demonstrate how model outputs can be transformed into decision-support information for cafeteria administrators.
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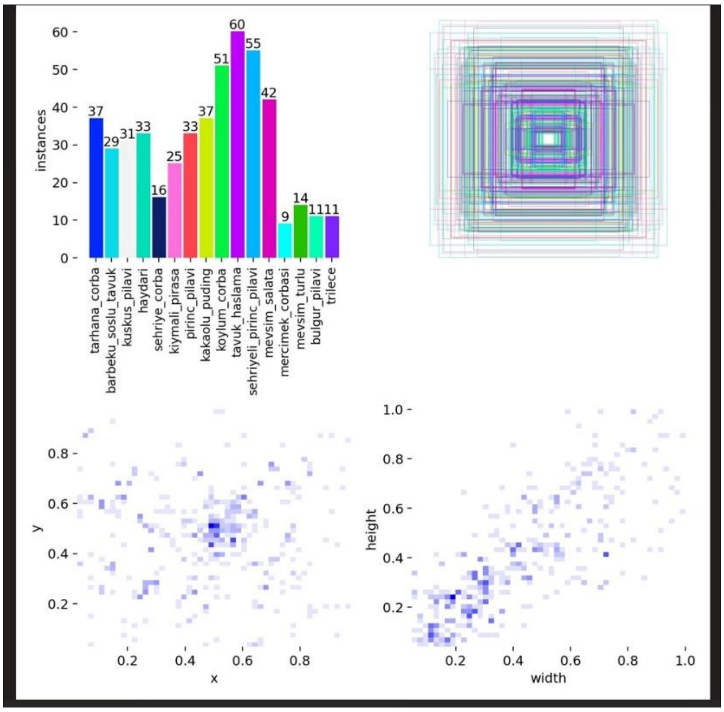
7. Presentation of Findings, Implementations, Prototypes, or Final Outputs

This section presents the preliminary findings, prototype implementations, and intermediate outputs obtained during the ongoing development of the project. Although the system is not yet fully finalized, the presented results demonstrate the feasibility and effectiveness of the proposed approach.

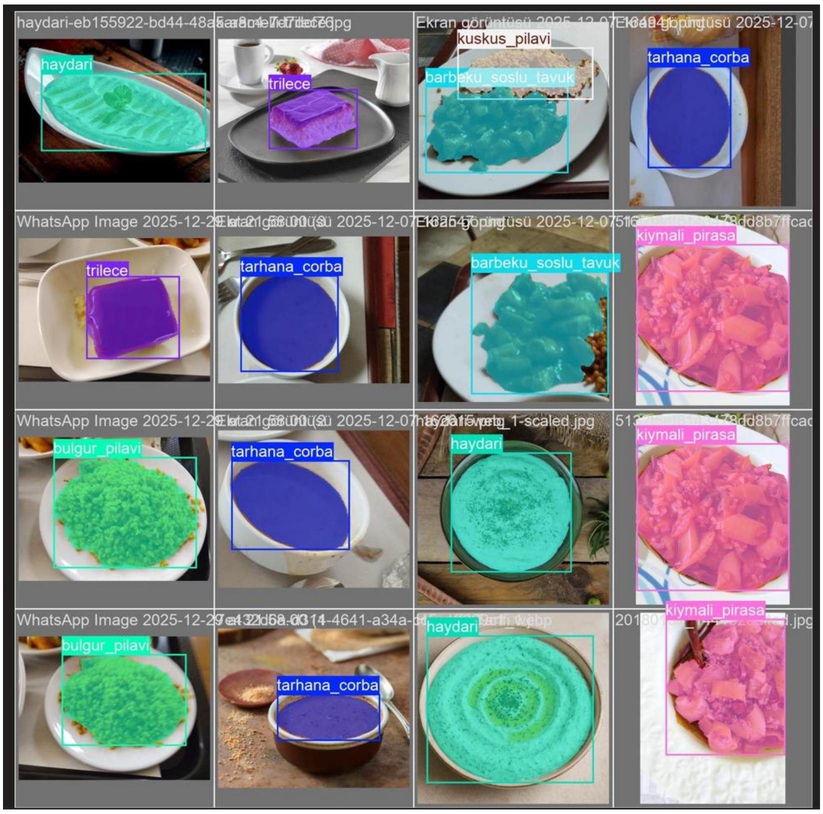
7.1 Dataset Characteristics and Class Distribution



The constructed dataset includes multiple dish categories commonly served in the university dining hall. Figure X illustrates the distribution of instances across dish classes. While some imbalance exists due to menu variability, each class contains a sufficient number of samples to support prototype-level model training and evaluation.

In addition, bounding box distribution analysis indicates that most food items appear near the center of images, reflecting consistent camera placement and tray alignment during data collection. This spatial consistency supports reliable object detection and simplifies model learning.

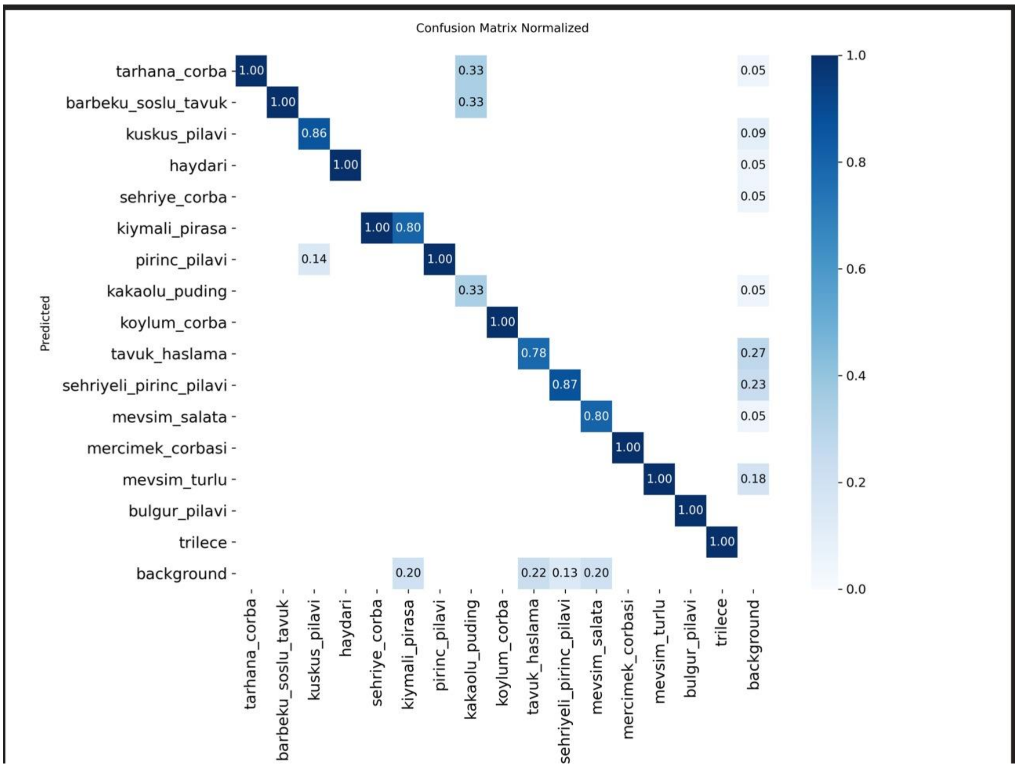
7.2 Object Detection and Localization Results



A YOLO-based object detection model was trained to identify and localize dish categories in food images. Figure Y presents qualitative detection results on sample images. The model successfully detects multiple dish types such as soups, pilaf varieties, desserts, and salads, assigning bounding boxes and class labels with high confidence.

These visual outputs confirm that the model can operate under realistic conditions, including variations in lighting, plate types, and food presentation. Although some bounding boxes are approximate, the detections are sufficient to support higher-level consumption analysis and system integration.

7.3 Classification Performance Analysis



To analyze class-level prediction behavior, a normalized confusion matrix is presented in Figure Z. High values along the diagonal indicate strong classification performance for several dish categories. Misclassifications are primarily observed between visually similar dishes, particularly among different pilaf types and mixed meals, which share similar textures and colors.

These results highlight both the strengths of the current model and the need for continued dataset expansion and refinement to improve discrimination between challenging classes.

7.4 F1–Confidence Curve Evaluation

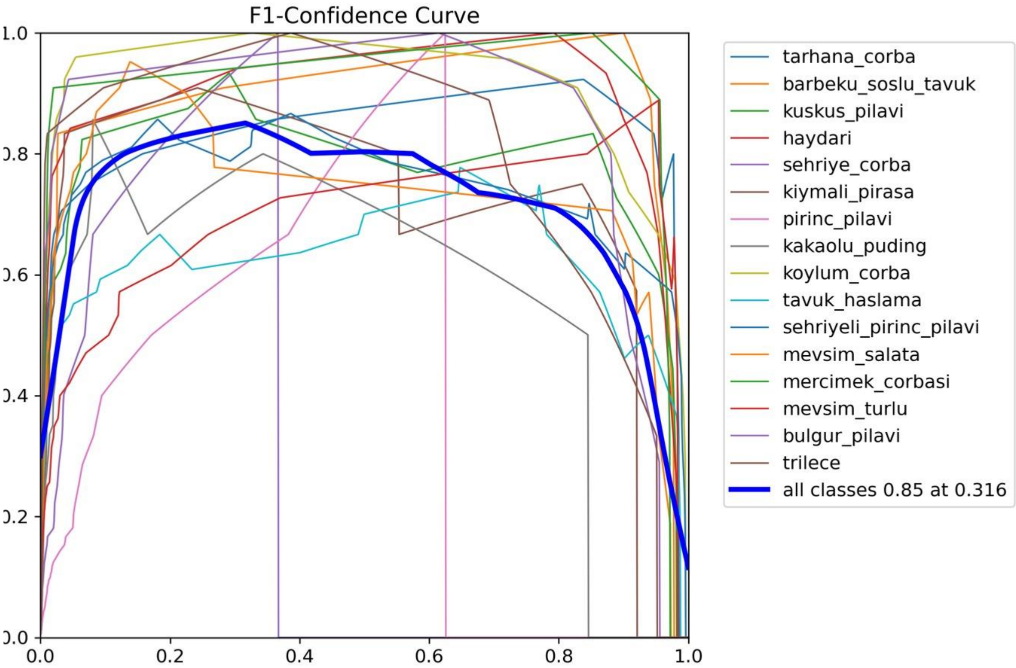


Figure W shows the F1–confidence curves for individual dish classes as well as the aggregated model performance across all classes. The aggregated F1–confidence curve demonstrates a balanced precision–recall trade-off at an experimentally selected confidence threshold.

The curves demonstrate that the model maintains stable performance across a wide range of confidence values, supporting its suitability for prototype-level deployment. Class-level variations reflect differences in dataset size and visual complexity.

7.5 Prototype-Level System Integration

The outputs presented in this section—including detection visualizations, confusion matrices, and confidence-based performance curves—represent intermediate yet functional components of the proposed system. Together, these results demonstrate that the core detection pipeline is operational and capable of producing meaningful outputs.

At the current stage, the system functions as a prototype that validates the feasibility of automated dish recognition and localization. Ongoing data collection and model training aim to further improve accuracy and robustness before final deployment.

8 Updated Gantt Chart

Task	Weeks	Status
Literature Review & Problem Definition	Week 1–2	Completed
System Design & Architecture Planning	Week 3	Completed
Dataset Strategy & Label Definition	Week 4	Completed
Initial Image Collection (Serving Trays)	Week 5–6	Completed
CVAT Setup & Annotation Workflow	Week 7	Completed
Dataset Annotation (Turkish Dishes – Phase 1)	Week 8–10	Completed
Initial YOLO Model Training	Week 10	Completed
Error Analysis & Misclassification Review	Week 11	Completed
Dataset Expansion (New Dishes)	Week 11–14	Ongoing
Iterative Model Fine-Tuning	Week 11–14	Ongoing
Performance Evaluation	Week 12–14	Ongoing
Final Dataset Completion	Future Work	Remaining
Final Model Optimization & Deployment	Future Work	Remaining

Due to the absence of an existing public dataset for Turkish cafeteria dishes, dataset construction and annotation are still ongoing. During this process, the model is continuously fine-tuned using newly annotated samples. This iterative training approach allows progressive performance improvement while the dataset is being expanded.

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9. Discussion of Challenges Encountered and How They Were Addressed

Throughout the project, several technical and practical challenges were encountered due to the complexity of real-world cafeteria environments and the ongoing nature of system development. This section summarizes the main challenges and the strategies applied to address them.

9.1 Dataset Imbalance and Limited Data Availability

Challenge:
The cafeteria menu varies daily, causing certain dish categories to appear less frequently in the dataset. This results in class imbalance and limits the number of samples available for some meals, which can negatively affect model performance.

Mitigation Strategy:
To address this issue, data collection is being conducted continuously to expand underrepresented classes. In addition, data augmentation techniques such as rotation, brightness adjustment, and contrast modification are applied to improve class balance and enhance model generalization.

9.2 Visual Similarity Between Dish Categories

Challenge:
Some dishes, particularly different pilaf varieties and mixed meals, share similar textures and colors. This visual similarity increases the likelihood of misclassification.

Mitigation Strategy:
The model training process is being refined through iterative evaluation using confusion matrix analysis. Difficult class pairs are identified, and additional samples are collected for these categories. Confidence threshold tuning is also used to reduce false positives.

9.3 Variability in Lighting and Image Conditions

Challenge:
Images are captured under varying lighting conditions and tray orientations, which can affect detection accuracy and bounding box precision.

Mitigation Strategy:
Basic preprocessing techniques such as illumination normalization and contrast enhancement are applied. Camera placement and capture angles are also monitored to maintain as much consistency as possible during data collection.

9.4 Real-Time Constraints and Model Selection

Challenge:
The system is intended for potential real-time or near-real-time deployment in a dining hall environment, requiring efficient inference without disrupting operations.

Mitigation Strategy:
A YOLO-based object detection model was selected due to its high inference speed and suitability for real-time applications. Lightweight model configurations are explored to balance detection accuracy and computational efficiency.

9.5 Ongoing Development and System Refinement

Challenge:
As the project is still under active development, some system components are not yet fully optimized.

Mitigation Strategy:
The project follows an iterative development approach. Dataset expansion, model retraining, and parameter tuning are continuously performed to improve system robustness and accuracy before final deployment.

10. Final Outcomes and Conclusions

This project investigates the feasibility of an AI-based system for analyzing food consumption and waste in a university dining hall environment. Although the project is still under active development, the work completed so far establishes a strong foundation for the final system.

At the current stage, a structured data acquisition strategy has been defined and implemented, and data collection is ongoing in real cafeteria conditions. A YOLO-based object detection model has been selected and integrated into the analysis pipeline, and preliminary training and evaluation results demonstrate that the model can successfully detect and localize multiple dish categories. These intermediate outputs confirm the technical feasibility of automated dish recognition under realistic environmental constraints.

In addition, a prototype-level system architecture has been designed, integrating data preprocessing, object detection, and result visualization components. Preliminary outputs such as detection visualizations, confusion matrices, and F1–confidence analyses provide meaningful insights into model behavior and guide further system refinement. While the model is not yet fully optimized, the current results indicate that the proposed approach is suitable for iterative improvement through additional data collection and training.

The remaining stages of the project focus on expanding the dataset, improving model robustness against visually similar dishes, and enhancing performance consistency across different lighting and tray conditions. Further training and evaluation will aim to increase detection accuracy and reduce misclassification rates. Ultimately, the final system is expected to support dish-level consumption analysis and provide actionable information for reducing food waste and supporting data-driven decision-making in institutional dining halls.

In conclusion, this project demonstrates that AI-driven food recognition and analysis is a viable approach for addressing food waste in university cafeterias. The outcomes achieved to date validate the proposed methodology and establish a clear path toward completing a functional and scalable system.

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